PH 101: STR and QM **Department of Physics**
Tutorial 3 **Indian Institute of Technology Guwahati**

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1. A particle of rest mass M having energy E (moving along x axis) decays into two particles of equal rest mass. In the lab frame, one of them is emitted along $+y$ axis. Consider the problem in x, y plane only. What are the energies of the created particles? Check your result with four momentum conservation also.

2. Consider the following one-dimensional situation, where all the motion is along the x axis. A particle has energy E' and momentum p' in frame S'. Frame S' moves at speed v with respect to frame S, in the positive $x = x'$ direction. What are E and p in S?

3. Derive the relativistic acceleration transformation
$$a'_x = \frac{a_x \left[1 - \frac{v^2}{c^2}\right]^{3/2}}{\left[1 - u_x v/c^2\right]^3},$$
where $a_x = du_x/dt$ and $a'_x = du'_x/dt'$.

4. The photoelectric effect suggests that light of frequency ν can be regarded as consisting of photons of energy $E = h\nu$, where $h = 6.63 \times 10^{-27}$ erg-sec.

   (a) Visible light has a wavelength in the range of 400-700 nm. What are the energy and frequency of a photon of visible light?
   (b) A standard microwave in our kitchen operates at roughly 2.5 GHz with a max power of 7.5 $\times 10^9$ erg/sec. How many photons per second can it emit?

5. In an experiment performed by Jonsson in 1961, electrons were accelerated through a 50 kV potential towards two slits separated by a distance d in order to perform double-slit experiment with electrons. Calculate the electron's de Broglie wavelength, λ.